

Customer Churn Analysis & Prediction

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Tools: Power BI, Power Query, DAX

Dataset: IBM Telco Customer Churn Dataset

Introduction

The objective of this project is to analyze customer churn patterns and identify the key factors influencing customer attrition. An interactive Power BI dashboard was created to support data-driven customer retention decisions.

Dataset Description

Total Records: 7032

Total Features: 21

Important columns include Gender, Contract, Payment Method, Internet Service, Monthly Charges, Total Charges and Churn.

Data Cleaning & Preprocessing

- Imported data into Power BI
- Corrected data types
- Removed 11 records with null values in TotalCharges
- Checked duplicates
- Created DAX measures for KPIs

Dashboard Overview

Insert the dashboard screenshot here.

Caption: Customer Churn Analysis Dashboard

Key KPIs

- Total Customers
- Churned Customers
- Active Customers
- Churn Rate (%)

Analysis

- Demographic Analysis
- Tenure Distribution
- Churn by Contract Type
- Churn by Payment Method
- Churn by Internet Service

Business Insights

- Month-to-month contracts show the highest churn.
- Electronic Check users churn more frequently.
- Fiber Optic customers have higher churn.
- Long-term contracts improve retention.

Recommendations

- Encourage long-term contracts.
- Improve support quality.
- Offer retention programs.
- Promote alternative payment methods.

Conclusion

The dashboard provides actionable insights that help identify high-risk customer segments and improve customer retention strategies.